



AREA II

DIAGNOSTIC EXAM

1. The peak runoff in the mainstream of a watershed has a mean of $128 \text{ m}^3/\text{s}$ and a standard deviation of $52 \text{ m}^3/\text{s}$. If the sample data is normally distributed, the magnitude of runoff corresponding to a probability of exceedance of 16% is

- a. $128 \text{ m}^3/\text{s}$
- b. $76 \text{ m}^3/\text{s}$
- c. $180 \text{ m}^3/\text{s}$
- d. $52 \text{ m}^3/\text{s}$

Gumbel Distribution

$$X_T = M + S(0.78 \ln T - 0.45)$$

where:

M = mean of the sample

S = standard deviation

T = return period

X_T = magnitude at a certain T

1. The peak runoff in the mainstream of a watershed has a mean of $128 \text{ m}^3/\text{s}$ and a standard deviation of $52 \text{ m}^3/\text{s}$. If the sample data is normally distributed, the magnitude of runoff corresponding to a probability of exceedance of 16% is

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b. $76 \text{ m}^3/\text{s}$

c. $180 \text{ m}^3/\text{s}$

d. $52 \text{ m}^3/\text{s}$

2. A culvert has been designed for a flood discharge with a return period of 50 years. The risk that the culvert will experience the design flood one or more times in 75 years is

a. 2%

b. 22%

c. 67%

d. 78%

e. 98 %

Basic Terms in Statistical Hydrology

- time period of interest (N)
- probability of occurrence (P)
- recurrence interval or return period (T)
the average number of years during which a storm of given magnitude (maximum depth or intensity) may be expected to occur once, *i.e.*, may be equaled or exceeded.

$$P = \frac{1}{T}$$

$$P = 1 - \frac{1}{T}$$



**Probability that a T-year storm may
occur in any series of N years**

$$P_{\text{exceedance}} = 1 - \left(1 - \frac{1}{T}\right)^N$$

**Probability that a T-year storm may
NOT occur in any series of N years**

$$P_{\text{non-exceedance}} = \left(1 - \frac{1}{T}\right)^N$$

2. A culvert has been designed for a flood discharge with a return period of 50 years. The risk that the culvert will experience the design flood one or more times in 75 years is

a. 2%

b. 22%

c. 67%

d. 78%

e. 98 %

3. The following infiltration capacity equation has been derived based on experimental data:

$$f = 10 + 5 e^{-12t}$$

where f is in mm/hr and t is in hr. The cumulative infiltration in mm after 45 minutes is

- a. 7.5mm
- b. 7.9 mm
- c. 79 mm
- d. 450 mm

Horton

$$f = fc + (fo - fc)e^{-kt}$$

f = instantaneous infiltration rate at time *t*

fc = final or constant infiltration rate

fo = initial infiltration rate

$$i = fct + \frac{(fo - fc)}{k} (1 - e^{-kt})$$



3. The following infiltration capacity equation has been derived based on experimental data: $f = 10 + 5e^{-12t}$

where f is in mm/hr and t is in hr. The cumulative infiltration in mm after 45 minutes is

a. 7.5mm

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c. 79 mm

d. 450 mm

4. The volume of water that can be stored in or released from an aquifer per unit horizontal area per unit change in hydraulic head is

- a. specific storage
- b. specific yield
- c. storage coefficient
- d. transmissivity

4. The volume of water that can be stored in or released from an aquifer per unit horizontal area per unit change in hydraulic head is

a. specific storage

b. specific yield – volume of water released from the storage in an unconfined aquifer per unit horizontal area per unit decline in water table.

c. storage coefficient

d. Transmissivity – an aquifer property which represents its ability to transmit water throughout its entire saturated thickness.

5. A fully penetrating well installed in a confined aquifer with a transmissivity of $197 \text{ m}^2/\text{day}$ is pumped at a constant rate. If the drawdowns in two observation wells 8 m and 22 m from the pumped well were observed to be 1.42 m and 0.44 m, respectively, the steady state pump discharge is

- a. 200 gpm
- b. 220 gpm
- c. 240 gpm
- d. 260 gpm
- e. 300 gpm

Governing Equations for Groundwater Flow

Steady Radial Flow in Confined Aquifers

$$Q = \frac{2\pi kb(h_2 - h_1)}{\ln\left(\frac{r_2}{r_1}\right)} \text{ or } Q = \frac{2\pi T(h_2 - h_1)}{\ln\left(\frac{r_2}{r_1}\right)}$$

Where:

Q = well discharge

k = hydraulic conductivity

b = average thickness of a confined aquifer

T = transmissivity or transmissibility, $T = kb$

h_2 = piezometric head at a radial distance r_2 from the well

h_1 = piezometric head at a radial distance r_1 from the well

r_1 and r_2 are radial distances corresponding to h_1 and h_2

Governing Equations for Groundwater Flow

Steady Radial Flow in **Unconfined Aquifers**

$$Q = \frac{\pi k (h_2^2 - h_1^2)}{\ln\left(\frac{r_2}{r_1}\right)}$$

Where:

Q = well discharge

k = hydraulic conductivity

h_2 = piezometric head at a radial distance r_2 from the well

h_1 = piezometric head at a radial distance r_1 from the well

r_1 and r_2 are radial distances corresponding to h_1 and h_2

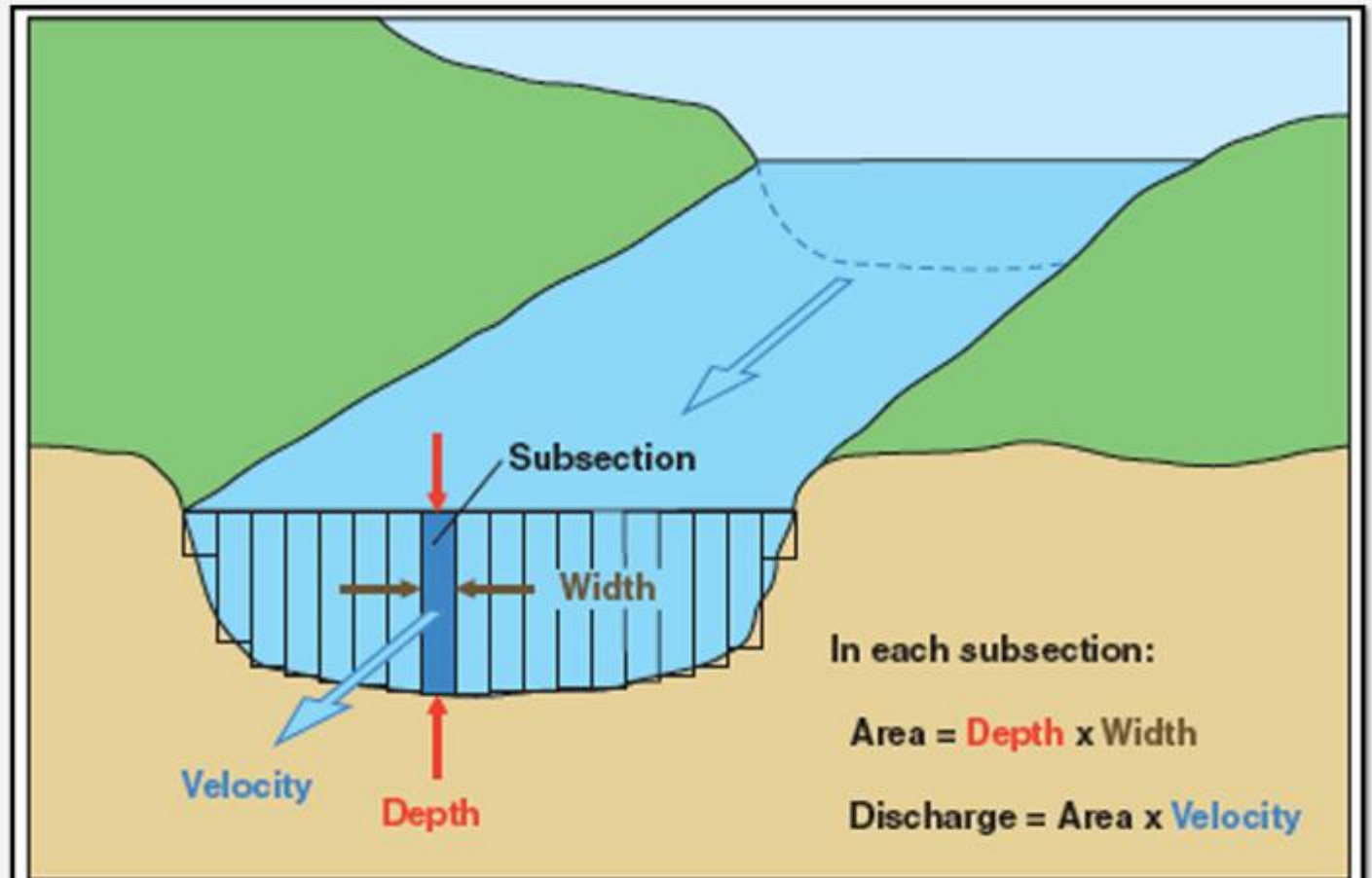
5. A fully penetrating well installed in a confined aquifer with a transmissivity of $197 \text{ m}^2/\text{day}$ is pumped at a constant rate. If the drawdowns in two observation wells 8 m and 22 m from the pumped well were observed to be 1.42 m and 0.44 m, respectively, the steady state pump discharge is

- a. 200 gpm
- b. 220 gpm**
- c. 240 gpm
- d. 260 gpm
- e. 300 gpm

6. The velocities at 0.2 and 0.8 of the depth of flow in a parabolic open channel are 1.0 m/s and 0.2 m/s, respectively. If the flow depth is 0.5 m and the top width is 1.5 m, the discharge is

- a. 500 lps
- b. 450 lps
- c. 300 lps
- d. 250 lps

A. Direct Method



Current-meter discharge measurements are made by determining the discharge in each subsection of a channel cross section and summing the subsection discharges to obtain a total discharge.

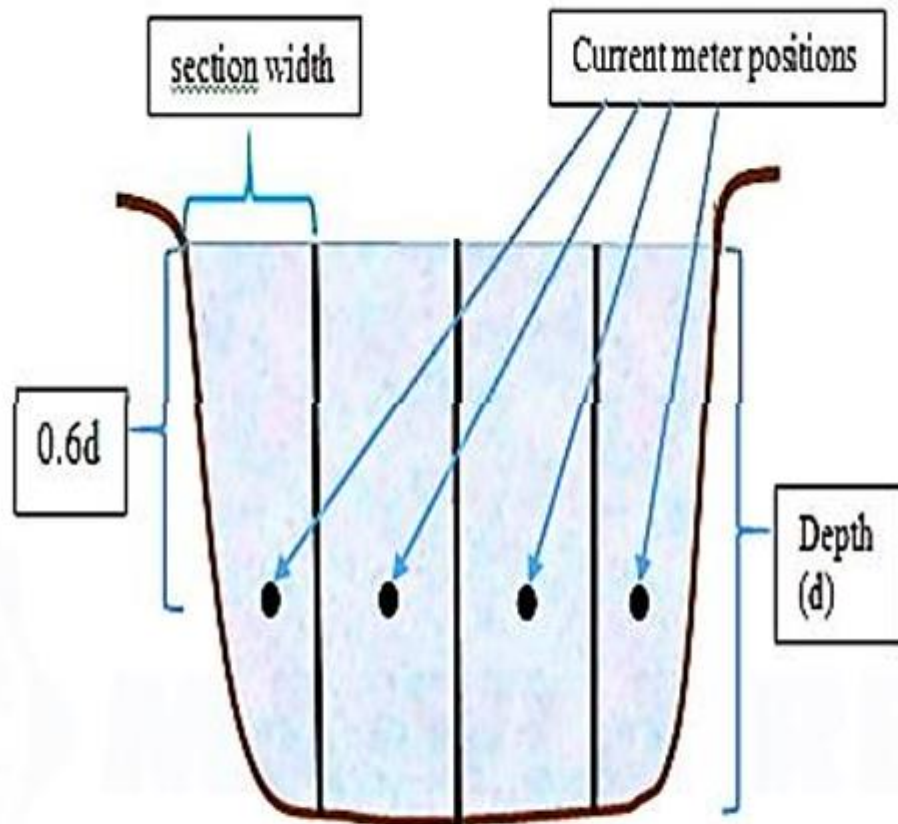


Figure 3-4. A canal cross section with established vertical sections and placements of the current meter for a less than 1 meter depth.

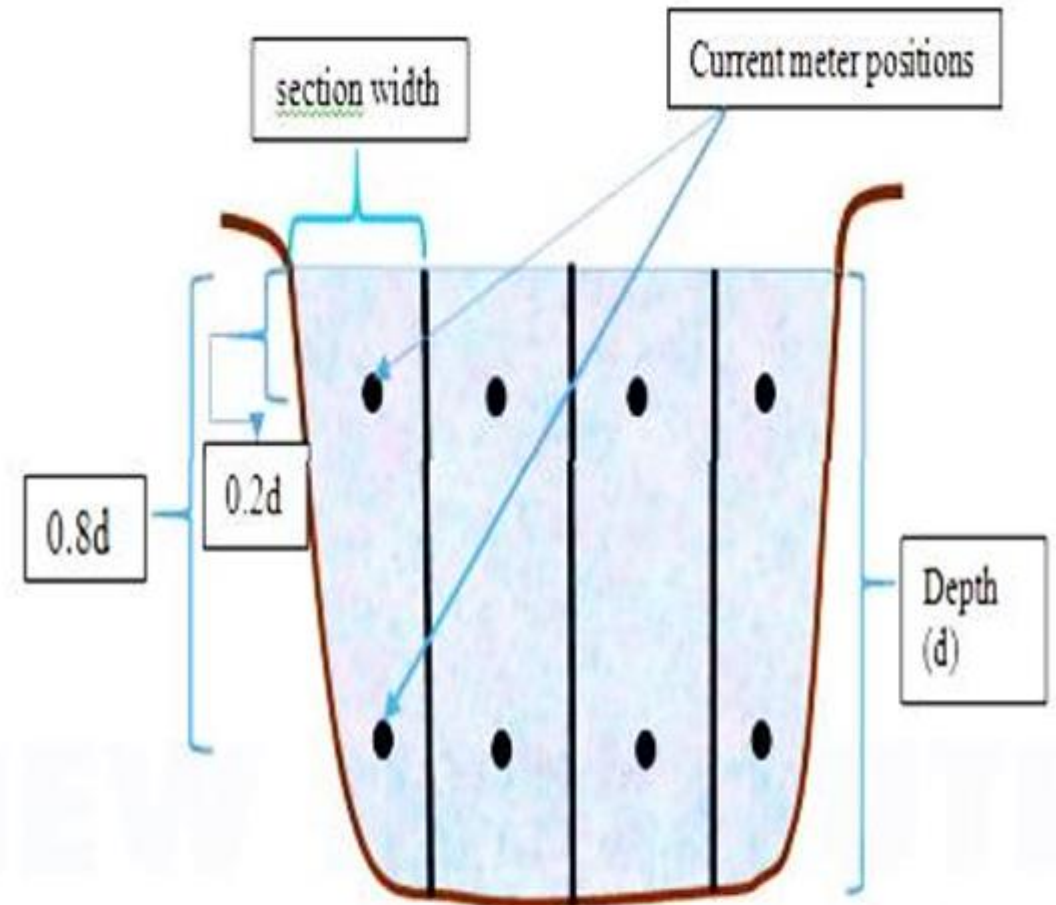


Figure 3-3. A canal cross section with established vertical sections and placements of the current meter for greater than 1 meter depth.

6. The velocities at 0.2 and 0.8 of the depth of flow in a parabolic open channel are 1.0 m/s and 0.2 m/s, respectively. If the flow depth is 0.5 m and the top width is 1.5 m, the discharge is

a. 500 lps

b. 450 lps

c. 300 lps

d. 250 lps

$$Q = AV$$

$$V = \frac{V_{0.2} + V_{0.8}}{2}$$

$$A_{\text{parabolic channel}} = \frac{2}{3}Ty$$

7. Given the following direct runoff hydrograph in m^3/s taken at 1-hr interval in the mainstream of a 50 sq. km watershed: 0, 2.8, 13.8, 44.8, 77.8, 37.8, 12.8, 4.5, 1.8, 1.3, 0.9, 0.6, 0.2. The equivalent depth of direct runoff is

a. 10.3 mm

b. 12.3 mm

c. 14.3 mm

d. 16.3 mm

7. Given the following direct runoff hydrograph in m^3/s taken at 1-hr interval in the mainstream of a 50 sq. km watershed: 0, 2.8, 13.8, 44.8, 77.8, 37.8, 12.8, 4.5, 1.8, 1.3, 0.9, 0.6, 0.2. The equivalent depth of direct runoff is

a. 10.3 mm

b. 12.3 mm

c. 14.3 mm

d. 16.3 mm

$$Q = \frac{Ad}{t}$$

8. The peak runoff in a 50 ha-watershed corresponding to a rainfall intensity of 25 mm/hr and a runoff coefficient of 0.8 is

a. $2.8 \text{ m}^3/\text{s}$

b. $28 \text{ m}^3/\text{s}$

c. $280 \text{ m}^3/\text{s}$

d. $1,000 \text{ m}^3/\text{s}$

8. The peak runoff in a 50 ha-watershed corresponding to a rainfall intensity of 25 mm/hr and a runoff coefficient of 0.8 is

a. 2.8 m^3/s

b. 28 m^3/s

c. 280 m^3/s

d. 1,000 m^3/s

9. Streamflow in a river was measured at various stages using a current meter. The results are given below:

Stage h (m)	Discharge (m ³ /s)
1.9	9.7
2.8	21.7
3.6	32.7
3.7	27.2
4.8	78.7
5.4	78.4
6.3	128.1
7.0	129.5
7.1	159.6
7.9	166.7

The stage-discharge relation of the form $q = ah^b$ is to be developed. Based on the observed data, the value of the exponent b is

a. 0.9

b. 1.9

c. 2.1

d. 2.9

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d. 2.9

10. The type of erosion that takes place in small channels which cannot be obliterated by tillage operations is called

a.sheet

b.rill

c.interrill

d.gully

10. The type of erosion that takes place in small channels which cannot be obliterated by tillage operations is called

- a. sheet
- b. rill
- c. interrill
- d. gully**



Rill Erosion



Splash Erosion



Gully Erosion



Sheet Erosion

11. The critical depth in a rectangular channel with a bottom width of 4.0 m for a discharge of 5 m³/s is

a. 0.34 m

b. 0.44 m

c. 0.54 m

d. 0.64 m

$$d_c = \sqrt[3]{\frac{q^2}{g}}$$

$$q = vd = \frac{Q}{b}$$

q = unit flow in m^3/s per meter width of canal
 Q = total flow in m^3/s
 B = channel width



11. The critical depth in a rectangular channel with a bottom width of 4.0 m for a discharge of 5 m³/s is

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b. 0.44 m

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12. A 30-inch diameter 1-meter long culvert has been designed to carry a maximum discharge of $1.7 \text{ m}^3/\text{s}$ when flowing half-full. Its roughness coefficient n is 0.014. If the inlet and the outlet are both unsubmerged, the difference in elevation between the inlet and the outlet should be

- a. 5 cm
- b. 7.5 cm
- c. 10 cm
- d. 12.5 cm
- e. 15 cm

Manning's Equation

$$V = \frac{k}{n} R^{2/3} S^{1/2}$$

***V** = velocity of flow (m/s or ft/s)*

***R** = hydraulic radius (m or ft)*

***k** = conversion factor (**1**-metric, **1.486**-english)*

***n** = roughness coefficient*

***S** = slope of energy grade line*

12. A 30-inch diameter 1-meter long culvert has been designed to carry a maximum discharge of $1.7 \text{ m}^3/\text{s}$ when flowing half-full. Its roughness coefficient n is 0.014. If the inlet and the outlet are both unsubmerged, the difference in elevation between the inlet and the outlet should be

a. 5 cm

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c. 10 cm

d. 12.5 cm

e. 15 cm

13. A concrete-lined trapezoidal canal is to carry a maximum discharge of $2.0 \text{ m}^3/\text{s}$. The required head loss over a horizontal distance of 1 km is 2.0 m. The side slope is to be set at 1:1 and the permissible b/d ratio is 3. If the Manning's roughness coefficient is 0.015, the design depth required is

- a. 0.42 m
- b. 0.56 m
- c. 0.64 m
- d. 0.78 m
- e. 1.1 m

Manning's Equation

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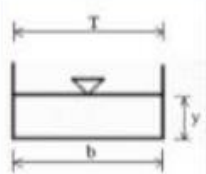
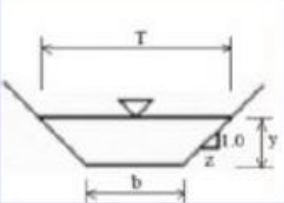
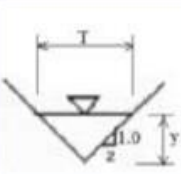
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CHANNEL GEOMETRY

	Flow Area (A)	Wetted perimeter (P)	Hydraulic Radius ($R=\frac{A}{P}$)	Top Width (T)	Hydraulic Depth ($D=\frac{A}{T}$)
	by	$b + 2y$	$\frac{by}{b + 2y}$	b	y
	$(b + zy)y$	$b + 2y\sqrt{1 + z^2}$	$\frac{(b + zy)y}{b + 2y\sqrt{1 + z^2}}$	$b + 2zy$	$\frac{(b + zy)y}{b + 2zy}$
	zy^2	$2y\sqrt{1 + z^2}$	$\frac{zy^2}{2y\sqrt{1 + z^2}}$	$2zy$	$\frac{1}{2}y$
	$\frac{2}{3}Ty$	$T + \frac{8y^2}{3T}$	$\frac{2T^2y}{2T^2 + 8y^2}$	$\frac{3A}{2y}$	$\frac{2}{3}y$



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a. 0.42 m

b. 0.56 m

c. 0.64 m

d. 0.78 m

e. 1.1 m

14. The most hydraulically efficient trapezoidal channel section to carry a discharge of $5 \text{ m}^3/\text{s}$ on a bed material with a roughness n of 0.02 and a bottom slope of 1 in 800 should have a bottom width of

a. 0.75 m

b. 1.35 m

c. 1.65 m

d. 2.25 m

e. 2.85 m

B. Best Hydraulic Section

For a constant AREA,

- Discharge is MAXIMUM
- Velocity is MAXIMUM

$$Q=VA$$

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

- Hydraulic Radius is MAXIMUM
- Perimeter is MINIMUM

$$R=A/P$$



Best Hydraulic Channel Section Properties

Cross sectional Shape	Area, A	Wetted Perimeter, P	Hydraulic Radius, R	Top Width, T
Rectangular (half of a square) $\theta = 90^\circ$	$2y^2$	$4y$	$0.5y$	$2y$
Trapezoidal (half of a hexagon) $\theta = 60^\circ$	$\sqrt{3}y^2$	$2\sqrt{3}y$	$0.5y$	$\frac{4\sqrt{3}}{3}y$
Triangular (half of a square on vertex) $\theta = 45^\circ$	y^2	$2\sqrt{2}y$	$\frac{\sqrt{2}}{4}y$	$2y$
Semi Circular	$\frac{\pi}{2}y^2$	πy	$0.5y$	$2y$



14. The most hydraulically efficient trapezoidal channel section to carry a discharge of 5 m³/s on a bed material with a roughness n of 0.02 and a bottom slope of 1 in 800 should have a bottom width of

- a. 0.75 m
- b. 1.35 m
- c. 1.65 m**
- d. 2.25 m
- e. 2.85 m

$$Q = (\sqrt{3}d^2) \times \frac{1}{n} (0.5d)^{2/3} S^{1/2}$$

$$5 = (\sqrt{3}d^2) \times \frac{1}{0.02} (0.5d)^{2/3} \left(\frac{1}{800} \right)^{1/2}$$

$$d = 1.429 \text{ m} \quad ; \quad A = \sqrt{3}d^2 = \sqrt{3}(1.429)^2 = 3.537 \text{ m}^2$$

Useful formulas:

$$Q = A \times \frac{1}{n} R^{2/3} S^{1/2}$$

$$A = bd + zd^2$$

Area for most efficient trapezoidal channel

$$A = \sqrt{3}d^2$$

$$3.537 = b(1.429) + (0.577)(1.429)^2$$

$$\mathbf{b = 1.65}$$

Hydraulic radius for most efficient trapezoidal channel

$$R = 0.5d$$

General formula for Area of a trapezoidal channel

$$A = bd + zd^2$$

15. The soil has a field capacity of 22% and a permanent wilting point of 12 %, both on a dry mass basis. The apparent specific gravity is 1.4. If the rootzone depth is 1.0 m and the management allowable deficit is 50%, the net application depth is

- a. 50 mm
- b. 60 mm
- c. 70 mm
- d. 80 mm

15. The soil has a field capacity of 22% and a permanent wilting point of 12 %, both on a dry mass basis. The apparent specific gravity is 1.4. If the rootzone depth is 1.0 m and the management allowable deficit is 50%, the net application depth is

a. 50 mm

b. 60 mm

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d. 80 mm

$$(FC - PWP) \times A_s \times MAD \times d_{rz}$$

16. A 3-ha of cornfield is to be irrigated at a discharge of 250 lps. The average available water holding capacity is 140 mm/m soil, the management allowable deficit is 40%, and the rootzone depth is 0.5 m. The reference crop potential evapotranspiration is 5 mm/day and the crop coefficient is 0.8. Assuming 70% application efficiency, the irrigation period should be

- a. 1 hr
- b. $1 \frac{1}{4}$ hrs
- c. $1 \frac{1}{3}$ hrs
- d. $1 \frac{1}{2}$ hrs

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- a. 1 hr
- b. 1 $\frac{1}{4}$ hrs
- c. 1 $\frac{1}{3}$ hrs**
- d. 1 $\frac{1}{2}$ hrs

$$\frac{\left(3ha \times \frac{10,000m^2}{ha} \times 0.5m\right) \times \frac{0.140m}{m} \times 0.4}{\left(250 \frac{L}{s} \times \frac{1m^3}{1000L} \times \frac{3600s}{1hr}\right) + \left[\left(\left(\frac{0.005m}{day} \times 0.8\right) \times \frac{1day}{24hrs}\right) \left(3ha \times \frac{10,000m^2}{ha}\right)\right] \times 0.70} \approx 1.33hrs$$

17. The soil has the following characteristics: porosity = 50 %, bulk density = 1.25g/cc, residual moisture content is 15% dry mass. The rooting depth is 300 mm and the duration of land soaking is 5 days. If the evaporation and percolation losses amount to 5 mm/day the gross land soaking requirement is

- a. 15.75 mm/day
- b. 17.75 mm/day
- c. 22.75 mm/day
- d. 27.75 mm/day

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- b. 17.75 mm/day
- c. 22.75 mm/day
- d. 27.75 mm/day**

$$LSR = [n - (RMC \times A_s)] \times d_{rz}$$

$$LSR = \frac{[0.50 - (0.15 \times 1.25)] \times 300mm}{5 \text{ days}} + 5 \frac{mm}{day}$$

18. A hydraulic structure that serves as an outlet of water in irrigation canals whereby water passes through and discharges into the main farm ditch of service area is called

- a.headgate
- b.checkgate
- c.turnout
- d.offtake
- e.inverted siphon

18. A hydraulic structure that serves as an outlet of water in irrigation canals whereby water passes through and discharges into the main farm ditch of service area is called

a. headgate

b. checkgate

c. turnout

d. offtake

e. inverted siphon

19. Sugarcane is to be irrigated using sprinkler irrigation. The sprinkler spacing is 12.2 m while the lateral spacing is 18.3 m. The soil water holding capacity is 120 mm/m, crop rooting depth is 1.0 m and management allowable deficit is 45%. If the irrigation period is 12 hrs and the application efficiency is 70%, the required sprinkler discharge is

- a. 0.2 lps
- b. 0.3 lps
- c. 0.4 lps
- d. 0.5 lps

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a. 0.2 lps

b. 0.3 lps

c. 0.4 lps

d. 0.5 lps

$$\frac{\left[(12.2m \times 18.3m \times 1m) \times \frac{1000L}{1m^3} \right] \times \frac{0.12m}{m} \times 0.45}{\left(12hrs \times \frac{3600s}{1hr} \right) \times 0.70}$$

20. Groundwater is to be pumped from a shallow aquifer for irrigation purposes. The required discharge is 300 gpm. The suction head is 8 ft and the maximum drawdown is 10 ft. The friction and minor losses are 0.45 ft and 0.3 ft, respectively. If the pump efficiency is 60%, the theoretical input power required is

- a. 2.0 hp
- b. 2.2 hp
- c. 2.4 hp
- d. 3.0 hp
- e. 3.2 hp

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d. 3.0 hp

e. 3.2 hp

$$P = \rho g Q H$$

$$P = \frac{QH}{3960}$$

21. Other terms for drip irrigation are _____irrigation and _____irrigation. This type of overhead irrigation operates at low pressure.

a.trickle - micro

b.sprinkle - slide

c.trickle - border

d.corrugation - trickle

21. Other terms for drip irrigation are _____irrigation and _____irrigation. This type of overhead irrigation operates at low pressure.

a.trickle - micro

b.sprinkle - slide

c.trickle - border

d.corrugation - trickle

22. These are usually rural traders or barrio buyers who do accumulation of the produce of individual farmers and bulk these for storage or for distribution to wholesalers

a.assemblers

b.wholesalers

c.retailers

d.agents and brokers

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a . assemblers

b . wholesalers

c . retailers

d . agents and brokers

23. This refers to wants for specific satisfiers backed up by an ability and willingness to buy them.

a.need

b.want

c.demand

d.value

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a. need

b. want

c. demand

d. value

24. This is a marketing function which facilitates transfer of ownership

- a.Assembly
- b.storage
- c.negotiation
- d.grading

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25. This is oftentimes not considered a marketing activity, but it is a marketing function. This is because agricultural products must pass through some levels of transformation before being made available to the users. This marketing function increases marketability of products by changing product forms.

- a. packaging/packing
- b. grading
- c. storage
- d. processing

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a. packaging/packing

b. grading

c. storage

d. processing

26. These are intermediaries who deal directly with the end-users or the consumers. They are the wholesalers' partners in distribution activities.

a.agents

b.retailers

c.assemblers

d.brokers

26. These are intermediaries who deal directly with the end-users or the consumers. They are the wholesalers' partners in distribution activities.

a. agents

b. retailers

c. assemblers

d. brokers

27. It is the state of felt deprivation
of some basic satisfaction

a.need

b.want

c.demand

d.exchange

27. It is the state of felt deprivation of some basic satisfaction

a . need

b . want

c . demand

d . exchange

28. It is anything offered for sale,
attention, and acquisition

a. Prospect

b. target market

c. product

d. market segment

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a. Prospect

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c. product

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29. Which is not part of the macro-environment?

a.demographic

b.suppliers of raw materials

c.technological

d.politico-legal

29. Which is not part of the macro-environment?

a. demographic

b. suppliers of raw materials

c. technological

d. politico-legal

30. This is a group of actual and potential buyers with similar needs and wants interacting with sellers offering various products or services to satisfy those needs and wants.

a. market

b. prospects

c. consumers

d. retailers

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a .market

b.prospects

c.consumers

d.retailers

31. The most dominant species cultured in brackishwater in the country are:

- a. Oyster and mussels
- b. Milkfish and shrimp
- c. Mudcrab and grouper
- d. Tilapia and milkfish

31. The most dominant species cultured in brackishwater in the country are:

a. Oyster and mussels

b. Milkfish and shrimp

c. Mudcrab and grouper

d. Tilapia and milkfish

32. Which of the following is not a major aquaculture species in the Philippines?

- a. Carp
- b. Salmon
- c. Seaweeds
- d. Tiger prawn

32. Which of the following is not a major aquaculture species in the Philippines?

a. Carp

b. Salmon

c. Seaweeds

d. Tiger prawn

33. This shrimp or prawn undergoes 11 larval stages to metamorphose to post larvae:

- a. Giant tiger prawn
- b. Whiteleg shrimp
- c. Giant freshwater prawn
- d. Indian shrimp

33. This shrimp or prawn undergoes 11 larval stages to metamorphose to post larvae:

a. Giant tiger prawn

b. Whiteleg shrimp

c. Giant freshwater prawn

d. Indian shrimp

34. Penaeid shrimps undergo these larval stages in sequence before they become post larvae:

a. Nauplius, megalopa, instar

b. Nauplius, zoea, mysis

c. Nauplius, mysis, zoea,

d. Zoea, megalopa, instar

34. Penaeid shrimps undergo these larval stages in sequence before they become post larvae:

a. Nauplius, megalopa, instar

b. Nauplius, zoea, mysis

c. Nauplius, mysis, zoea,

d. Zoea, megalopa, instar

35. The Philippines ranks as the world's third largest producer of:

- a. Shrimp
- b. Milkfish
- c. Sea cucumber
- d. Seaweeds

35. The Philippines ranks as the world's third largest producer of:

a. Shrimp

b. Milkfish

c. Sea cucumber

d. Seaweeds

36. Water like in estuarine areas with intermediate salinity due to the mixing of fresh and sea water is referred to as:

- a. Freshwater
- b. Seawater
- c. Brackishwater
- d. Marine water

36. Water like in estuarine areas with intermediate salinity due to the mixing of fresh and sea water is referred to as:

- a. Freshwater
- b. Seawater
- c. Brackishwater**
- d. Marine water

37. Breeding and growout production of commercially important marine species are grown in zonified marine cages which the Bureau of Fisheries and Aquatic Resources refers to as:

- a. Industrial cage parks
- b. Seawater cage farms
- c. Ocean cage parks
- d. Mariculture parks

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- a. Industrial cage parks
- b. Seawater cage farms
- c. Ocean cage parks
- d. Mariculture parks**

38. Eye ablation is a way of inducing females to spawn in:

- a. Milkfish
- b. Mussels
- c. Oysters
- d. Shrimp

38. Eye ablation is a way of inducing females to spawn in:

a. Milkfish

b. Mussels

c. Oysters

d. Shrimp

39. A piece of wire 120 meters long is bent to form a right triangle with a hypotenuse 50 meters long. Find the length of the other two sides of the triangle.

a. 30 m and 40 m

b. 35 meters

c. 50 m and 20 m

d. 45m and 25m

39. A piece of wire 120 meters long is bent to form a right triangle with a hypotenuse 50 meters long. Find the length of the other two sides of the triangle.

a. 30 m and 40 m

b. 35 meters

c. 50 m and 20 m

d. 45m and 25m

$$a^2 + b^2 = c^2$$

40. After 2 hours of driving, Noel and Chester are 260 meters apart. Find their average speed if their difference in speed is 30 km per hour (kph).

a. 100 and 130 kph

b. 50 and 80 kph

c. 80 and 110 kph

d. 30 and 60 kph

40. After 2 hours of driving, Noel and Chester are 260 meters apart. Find their average speed if their difference in speed is 30 km per hour (kph).

a. 100 and 130 kph

b. 50 and 80 kph

c. 80 and 110 kph

d. 30 and 60 kph

41. Kim can do the job in 6 days but if Andre will help they can finish it in 2 days. How long will it take Andre to do the job alone?

a. 5 days

b. 4 days

c. 3 days

d. 2 days

41. Kim can do the job in 6 days but if Andre will help they can finish it in 2 days. How long will it take Andre to do the job alone?

a. 5 days

b. 4 days

c. 3 days

d. 2 days

$$\frac{1}{\frac{1}{6} + \frac{1}{\text{Andre}}} = 2$$

42. It is one of the realities of marketing. It includes actual and potential rivals in product and service marketing

a.competition

b.marketing environment

c.supply chain

d.marketing channels

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a . competition

b . marketing environment

c . supply chain

d . marketing channels

43. The key objectives of each of the 3 Cs of marketing must be attained to be considered

a. Profitable

b. effective

c. marketing-oriented

d. successful

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a. Profitable

b. effective

c. marketing-oriented

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44. This is the set of tools a firm uses to pursue the marketing objectives for the target market

a.marketing environment

b.marketing mix

c.marketing channels

d.relationship marketing

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45. This is an indispensable component of a firm to continuously satisfy customers. It comes from having excess of sales over cost and expenses

a.sales

b.profit

c.revenues

d.market share

45. This is an indispensable component of a firm to continuously satisfy customers. It comes from having excess of sales over cost and expenses

a. sales

b. profit

c. revenues

d. market share

46. _____ are products that underwent secondary or the second level of transformation. These are in the forms which are for use by other industries and processors but may not be consumed.

- a. fresh agricultural products
- b. finished products
- c. semi-processed products
- d. processed products

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a. fresh agricultural products

b. finished products

c. semi-processed products

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47. These are activities that create faster and more efficient performance of other marketing activities. These include market intelligence and marketing information dissemination, financing, provision of marketing support facilities, and promotions.

- a. transactional activities
- b. physical handling and distribution of products
- c. facilitating activities
- d. agroservices activities

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a. transactional activities

b. physical handling and distribution of products

c. facilitating activities

d. agroservices activities

48. This is a continuous function of marketing. The objective of this function is to inform prospective buyers of product availability and also to increase sales.

a.pricing

b.promotion

c.research and development

d.distribution

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49. Because of the variability in agricultural outputs, _____ must be done to facilitate buying, selling, transportation, storage, and pricing of the produce. This is also done to standardize measurements, which could be in terms of size, weight, and overall quality.

a. grading

b. assembly

c. negotiation

d. packing/packaging

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c. negotiation

d. packing/packaging

50. These are the types of middlemen who do the business of negotiating in behalf of the producers and the buyers. They do not assume ownership of the products that they are handling.

a. Retailers

b. wholesalers

c. assemblers

d. agents and brokers

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a. Retailers

b. wholesalers

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d. agents and brokers

51. This is a fairly homogeneous group of customers who will respond to marketing mix in a similar way

a.market potential

b.market segment

c.market prospects

d.target market

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52. This is the proportion of a company's total sales of a product during a defined duration in a specific market or geographical area

a.primary demand

b.market share

c.market potential

d.profit

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a.primary demand

b.market share

c.market potential

d.profit

53. This is the consumer's estimate of the overall capacity of the product to meet desired needs and wants

a.satisfaction

b.delight

c.quality

d.value

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a.satisfaction

b.delight

c.quality

d.value

54. This describes the long process which starts from raw material sourcing to the marketing of the final products

a.supply chain

b.marketing mix

c.marketing channels

d.transactions

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a. supply chain

b. marketing mix

c. marketing channels

d. transactions

55. Marketing overcomes several separations. One of these is _____. This happens when producers tend to locate where it is economical to produce while consumers are located in many scattered locations.

- a. Separation in Time
- b. Separation in Information
- c. Separation in location
- d. Separation in value

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- b. Separation in Information
- c. Separation in location**
- d. Separation in value

56. Value can be increased by:

a.raising benefits

b.reducing costs

c.raising benefits by more than the
raise in costs

d.all of the above

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b.reducing costs

c.raising benefits by more than the
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57. The value chain consists of primary activities and support activities. An example of a support activity is:

- a. procurement
- b. inbound logistics
- c. operations
- d. all of the above

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a .procurement

b.inbound logistics

c.operations

d.all of the above

58. This is done to “homogenize”
markets

a.prospecting

b.target marketing

c.exchange

d.segmentation

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markets

a.prospecting

b.target marketing

c.exchange

d.segmentation

59. In which of the following is the word STATISTICS used in the singular sense?

I. The *statistics* on teen-age pregnancy bring concerns to parents of UPLB students.

II. The different *statistics* courses offered by the INSTAT are useful to most students who are going to conduct research in the future.

a. I only

b. II only

c. Both I and II

d. Neither I nor II

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c. Both I and II

d. Neither I nor II

60. Which of the following made use of *inferential statistics*?

I. According to the survey conducted by Pulse Asia Inc. using randomly selected respondents, thirteen party-groups would have won at least one seat in Congress had the elections been held during the first week of the month.

II. Based on the complete list of voters for District 5 of Manila, 300 eligible voters have exactly the same name.

- a. I only
- b. II only
- c. Both I and II
- d. Neither I nor II

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I. According to the survey conducted by Pulse Asia Inc. using randomly selected respondents, thirteen party-groups **would have won at least one seat** in Congress had the elections been held during the first week of the month.

II. Based on the complete list of voters for District 5 of Manila, 300 eligible voters have exactly the same name.

a. I only

b. II only

c. Both I and II

d. Neither I nor II

For numbers 61 to 64:

To characterize the habits of UPLB students enrolled for the school year 2007-08 in using the library, the following information were asked from each student:

F -frequency of using the library (Everyday, Thrice-a-week, Once-a-week , Never)

H -average number of hours spent in the library per week

S -library section most frequently visited (Reserve, Circulation, Periodicals)

I -IQ score

Y -student classification (1 - Freshman, 2 - Sophomore, 3 - Junior, 4 - Senior, 5 - Graduate)

N -number of books borrowed this semester

M -major field of specialization

61. The universe in the study is the set of ____.

a. all UPLB students

b. all UPLB students enrolled for the school year 2007-08

c. all freshmen students enrolled for the school year 2007-08

d. students who have visited the library at least once

61. The universe in the study is the set of ____.

a. all UPLB students

b. all UPLB students enrolled for the school year 2007-08

c. all freshmen students enrolled for the school year 2007-08

d. students who have visited the library at least once

62. A possible population(s) is(are) the set of__.

- a. values for student classification of UPLB students enrolled for 2007-08
- b. library sections most frequently visited by UPLB students enrolled for 2007-08
- c. number of books borrowed this semester by UPLB students enrolled for 2007-08
- d. major fields of specialization of UPLB students enrolled for 2007-08

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- a. values for student classification of UPLB students enrolled for 2007-08
- b. library sections most frequently visited by UPLB students enrolled for 2007-08
- c. number of books borrowed this semester by UPLB students enrolled for 2007-08
- d. major fields of specialization of UPLB students enrolled for 2007-08

63. Which of the following is(are) TRUE?

I. The variables H, I and Y are quantitative and are of the ratio, interval and nominal levels, respectively.

II. The variables S and F are qualitative variables and are of the nominal and ordinal levels, respectively.

a. I only

b. II only

c. Both I and II

d. Neither I nor II

63. Which of the following is(are) TRUE?

I. The variables H, I and Y are quantitative and are of the ratio, interval and nominal levels, respectively.

II. The variables S and F are qualitative variables and are of the nominal and ordinal levels, respectively.

a. I only

b. II only

c. Both I and II

d. Neither I nor II

64. Which of the following is(are) TRUE?

I. The variables N and Y are both discrete quantitative variables.

II. The variable I is a quantitative variable in the interval scale.

a. I only

b. II only

c. Both I and II

d. Neither I nor II

64. Which of the following is(are) TRUE?

I. The variables N and Y are both discrete quantitative variables.

II. The variable I is a quantitative variable in the interval scale.

a. I only

b. II only

c. Both I and II

d. Neither I nor II

65. The most appropriate method to use in order to present the trend of the performance of Maquiling School graduates in the UP Rural Entrance Exam over the last 25 years is__.

a.textual

b.qualitative FDT

c.graphical

d.quantitative FDT

65. The most appropriate method to use in order to present the trend of the performance of Maquiling School graduates in the UP Rural Entrance Exam over the last 25 years is__.

a.textual

b.qualitative FDT

c.graphical

d.quantitative FDT

69. Consider the table below.

Table 1. Partial FDT on the net sales (in million pesos) of 24 manufacturing firms in the country.

<u>Net Sales</u>	<u>Relative Frequency(5)</u>
1 - 3	12.5
4 - 6	14.5
7 - 9	40.0
10 - 12	22.5
13 and above	10.5

How many firms earned less than P4 million ?

a . 12.5

b . 6

c . 3

d . 3.48

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13 and above	10.5

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a . 12.5

b . 6

c . 3

d . 3.48

67. Consider the frequency distribution table below of the number of text messages Anne received per day last September.

CLASSES	FREQ	<RCF (%)	>RCF (%)
0-5	1	3.33	100.00
6-11	5	20.00	96.67
12-17	4	33.33	80.00
18-23	11	70.00	66.67
24-29	8	96.67	30.00
30-35	1	100.00	33.33

Which of the following is(are) TRUE?

- a. About 96.67% of the number of text messages Anne received per day last September is greater than six and less than 29.
- b. About 33.33% of the number of text messages Anne received per day last September is greater than 12.
- c. Twenty percent of the number of text messages Anne received per day last September ranges from 0 to 11.
- d. Thirty percent of the number of text messages Anne received per day last September ranges from 24 to 29.

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CLASSES	FREQ	<RCF (%)	>RCF (%)
0-5	1	3.33	100.00
6-11	5	20.00	96.67
12-17	4	33.33	80.00
18-23	11	70.00	66.67
24-29	8	96.67	30.00
30-35	1	100.00	33.33

Which of the following is(are) TRUE?

- a. About 96.67% of the number of text messages Anne received per day last September is greater than six and less than 29.
- b. About 33.33% of the number of text messages Anne received per day last September is greater than 12.
- c. Twenty percent of the number of text messages Anne received per day last September ranges from 0 to 11.**
- d. Thirty percent of the number of text messages Anne received per day last September ranges from 24 to 29.

For numbers 71 to 73:

The following summarizes the amount of soft drinks consumed per month (ounces) by 200 residents of Brgy. San Lorenzo, San Pablo City:

AMOUNT CONSUMED	32	85	130	150	375
NO. OF RESIDENTS	50	45	70	34	1

68. The average amount of soft drinks consumed per month by residents of Brgy. San Lorenzo, San Pablo City is ____.

a. 70

b. 100

c. 130

d. 156

68. The average amount of soft drinks consumed per month by residents of Brgy. San Lorenzo, San Pablo City is ____.

a. 70

b. 100

c. 130

d. 156

69. Seventy five percent of the residents consumed at most _____ amount of soft drinks per month (ounces) .

a. 70

b. 100

c. 130

d. 150

69. Seventy five percent of the residents consumed at most _____ amount of soft drinks per month (ounces).

a. 70

b. 100

c. 130

d. 150

70. The most appropriate measure of central tendency is_____.

a.maximum

b.mean

c.median

d.mode

70. The most appropriate measure of central tendency is_____.

a.maximum

b.mean

c.median

d.mode

71. The mean height of this year's batch of graduating Prep students in a private school is 88 cm. Two students are extremely tall when compared to their classmates. The height distribution of this batch of students is_____.

- a. symmetric
- b. positively skewed
- c. normal
- d. negatively skewed

71. The mean height of this year's batch of graduating Prep students in a private school is 88 cm. Two students are extremely tall when compared to their classmates. The height distribution of this batch of students is_____.

a. symmetric

b. positively skewed

c. normal

d. negatively skewed

72. It is the environmentally and genetically determined observable appearance of an organism; the expression of genes which can be measured by our senses. What we see in an animal for some trait.

- a. genotype
- b. phenotype
- c. generation
- d. pedigree

72. It is the environmentally and genetically determined observable appearance of an organism; the expression of genes which can be measured by our senses. What we see in an animal for some trait.

a. genotype

b. phenotype

c. generation

d. pedigree

73. A term that is used to explain the fact that the hybrids or crossbreed are usually better and more vigorous than their parents.

- a.heterosis or hybrid vigor
- b.halitosis or hybrid superiority
- c.heserosis or hybrid superiority
- d.hesterosis or hybrid vigor

73. A term that is used to explain the fact that the hybrids or crossbreeds are usually better and more vigorous than their parents.

a. heterosis or hybrid vigor

b. halitosis or hybrid superiority

c. heserosis or hybrid superiority

d. hesterosis or hybrid vigor

74. Period during which the
reproductive organs first become
functional

- a.heredity
- b.puberty
- c.adulthood
- d.maturity

74. Period during which the reproductive organs first become functional

a. Heredity – genetic transmission of traits from parents to offspring

b. puberty

c. adulthood

d. Maturity – to reaching a state of full development

75. The carcass or carcass parts coming from animals sufficiently mature and healthy.

a.filly

b.colt

c.chevon

d.meat

75. The carcass or carcass parts coming from animals sufficiently mature and healthy.

a.filly

b.colt

c.chevon

d.meat

76. Meat that is derived from cattle that is less than one year old, or meat from calf.

a.hot meat

b.Venison

c.veal

d.mutton

76. Meat that is derived from cattle that is less than one year old, or meat from calf.

a.hot meat – freshly slaughtered meat

b.Venison – Meat from deer

c.veal

d.Mutton – Meat from sheep

Meat Terms

- **Meat** - the carcass or carcass parts coming from animals sufficiently mature and healthy.
- **Beef** - meat from **cattle more than a year old**.
- **Veal** - meat from **cattle less than one year old**; meat from calf.
- **Carabeef** - meat from **carabaos more than one year old**.
- **Chevon** - meat from **goats**.
- **Pork** - meat from **swine**.
- **Mutton** - meat from **sheep**.
- **Lamb** - meat from **young sheep**.
- **Venison** - meat from **deers**.
- **Hot meat** - meat from an illegal source.
- **Fresh meat** - meat that has not undergone any substantial change from the time of slaughter.
- **Meat by-products** - edible by-products from the slaughter house.

77. Glands that secrete chemical mediators called hormones that regulate growth and development, metabolism, reproduction, stress responses, body and tissue fluid, and electrolyte balance.

- a.mammary glands
- b.endocrine glands
- c.resistive glands
- d.maturity glands

77. Glands that secrete chemical mediators called hormones that regulate growth and development, metabolism, reproduction, stress responses, body and tissue fluid, and electrolyte balance.

a. mammary glands – responsible for producing milk in mammals

b. endocrine glands

c. resistive glands

d. maturity glands

78. The pigs farrowed by a sow or the pups whelped by a bitch at one delivery period; also the accumulation of materials used for bedding of farm animals.

a. trash

b. litter

c. pool

d. crib

78. The pigs farrowed by a sow or the pups whelped by a bitch at one delivery period; also the accumulation of materials used for bedding of farm animals.

a. trash

b. litter

c. pool

d. crib

79. The period from conception to birth of the young

- a. Mastication
- b. Lactation
- c. Fecundation
- d. gestation

79. The period from conception to birth of the young

- a. Mastication
- b. Lactation
- c. Fecundation
- d. gestation**

80. Impregnation of fertilization

a. Mastication

b. Lactation

c. Fecundation

d. gestation

80. Impregnation of fertilization

- a. Mastication – process of **chewing food**
- b. Lactation – secretion of **milk** from the mammary glands after birth.
- c. Fecundation**
- d. gestation

81. The hormone that causes ovulation of the reproductive tract in birds is called

- a. luteinizing hormone
- b. growth hormone
- c. mammary hormone
- d. gestation hormone

81. The hormone that causes ovulation of the reproductive tract in birds is called

a.luteinizing hormone

b.growth hormone – Primarily regulates growth and development.

c.mammary hormone – Involved in milk production in mammals

d.gestation hormone

82. Metabolic process by which complex substances are converted by living cells into more simple compounds

a.catabolism

b.anabolism

c.Homogenism

d.metabolism

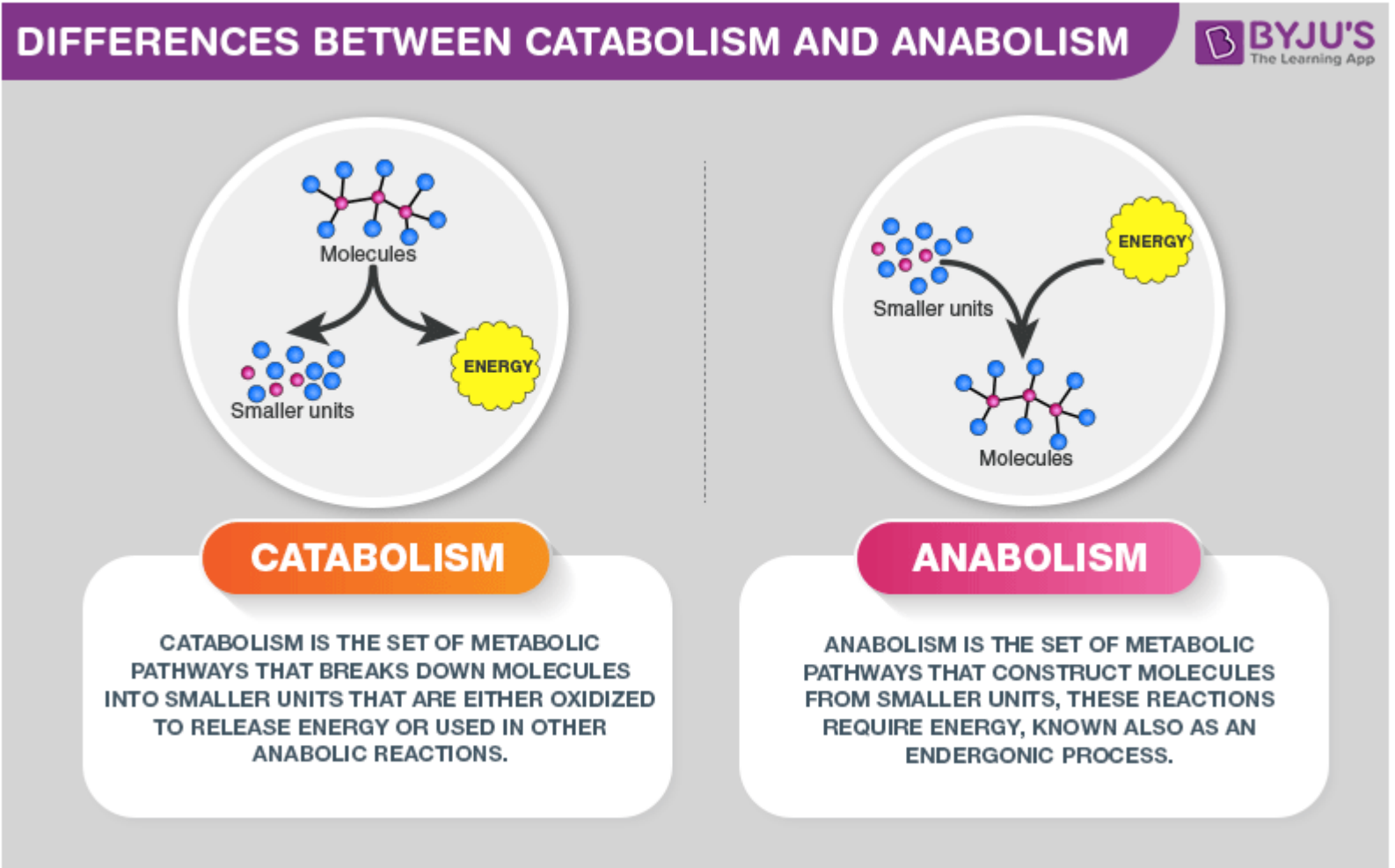
82. Metabolic process by which complex substances are converted by living cells into more simple compounds

a. catabolism

b. anabolism

c. Homogenism

d. metabolism



83. The term that pertains to the end product of decomposition

a.slime

b.sludge

c.humus

d.organic matter

83. The term that pertains to the end product of decomposition

a. slime

b. sludge

c. humus

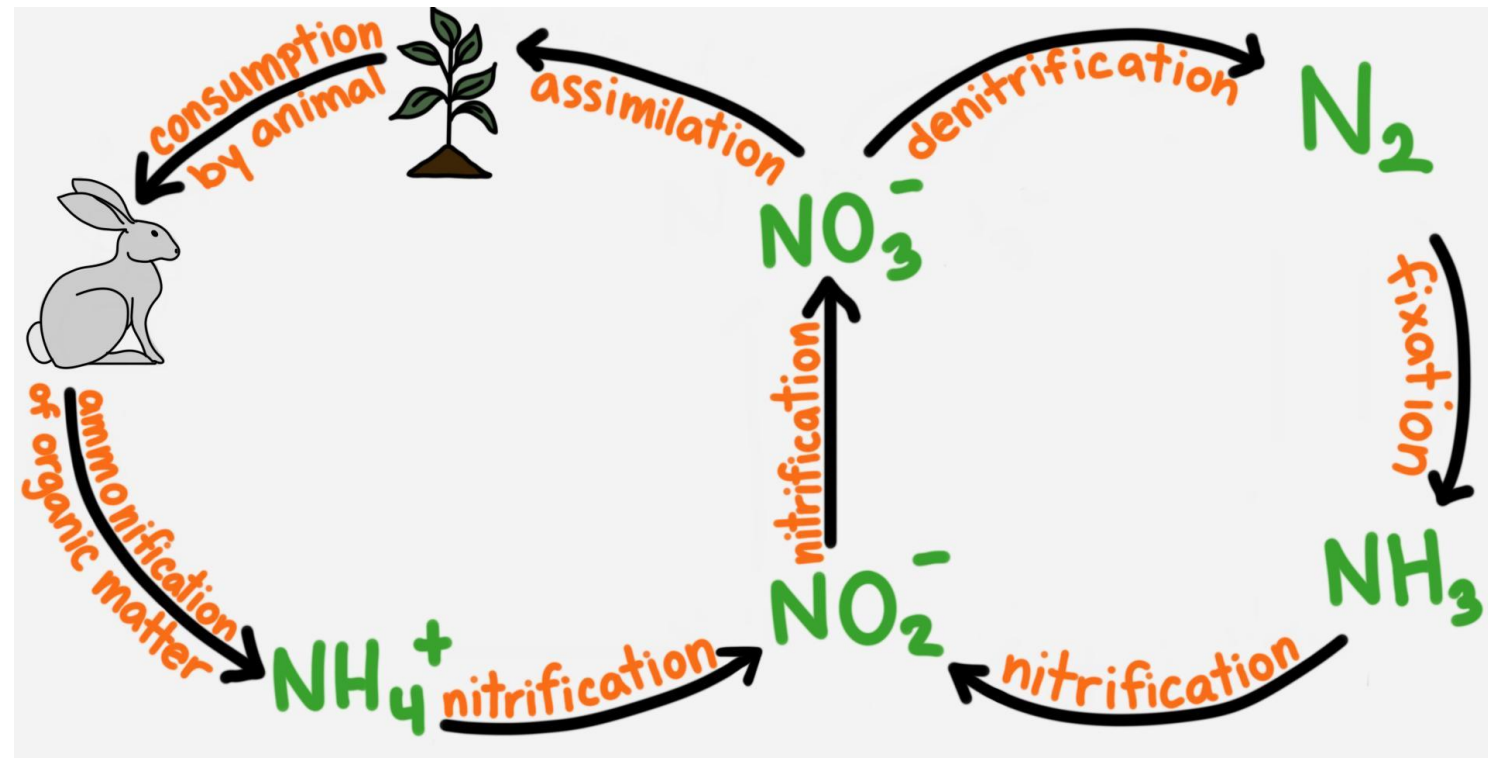
d. organic matter

84. The process of converting soil Nitrate into gaseous Nitrogen or Nitrous Oxide.

- a.ammonification
- b.nitrification
- c.denitrification
- d.siltation

87. The process of converting soil Nitrate into gaseous Nitrogen or Nitrous Oxide.

- a. ammonification
- b. nitrification
- c. denitrification**
- d. siltation



88. Water which has been absorbed from an atmosphere of water vapor as a result of attractive forces in the surface of the particles.

- a. groundwater
- b. hygroscopic water
- c. capillary water
- d. gravitational water

88. Water which has been absorbed from an atmosphere of water vapor as a result of attractive forces in the surface of the particles.

a. groundwater

b. hygroscopic water

c. capillary water

d. gravitational water

89. Water held by the surface tension forces as a continuous film around the particles and in the capillary spaces

a. groundwater

b. hygroscopic water

c. capillary water

d. gravitational water

89. Water held by the surface tension forces as a continuous film around the particles and in the capillary spaces

a. groundwater

b. hygroscopic water

c. capillary water

d. gravitational water

87. Water that is not held by soil but drains under the influence of gravity

a. groundwater

b. hygroscopic water

c. capillary water

d. gravitational water

87. Water that is not held by soil but drains under the influence of gravity

a. groundwater

b. hygroscopic water

c. capillary water

d. gravitational water

88. _____ are remains of plants and
animals in varying stages of
decomposition

a.humus

b.organic matter

c.slime

d.sludge

88. _____ are remains of plants and
animals in varying stages of
decomposition

a.humus

b.organic matter

c.slime

d.sludge

89. A flower with all the parts of the perianth and pistil and stamen.

a. perfect flower

b. complete flower

c. complete & perfect flower

d. fertile flower

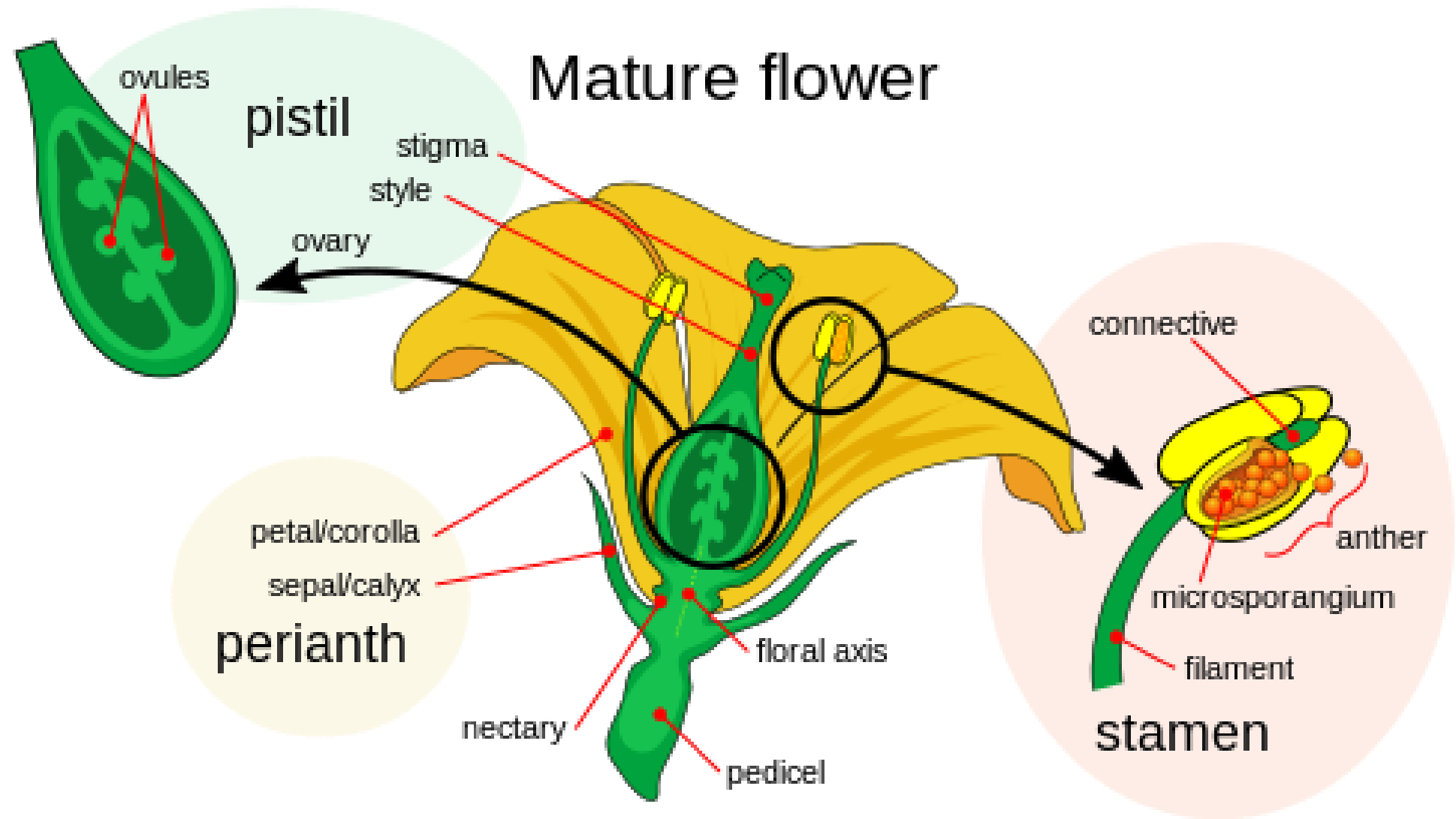
89. A flower with all the parts of the perianth and pistil and stamen.

a. perfect flower

b. complete flower

c. complete & perfect flower

d. fertile flower



90. It pertains to the starchy interior of a grain.

a. endosperm

b. endocarp

c. epicotyls

d. ecotype

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91. _____ is the vertical section from the surface downward to the bedrock, individual layers are regarded as horizons.

- a. field profile
- b. soil profile
- c. water profile
- d. plant profile

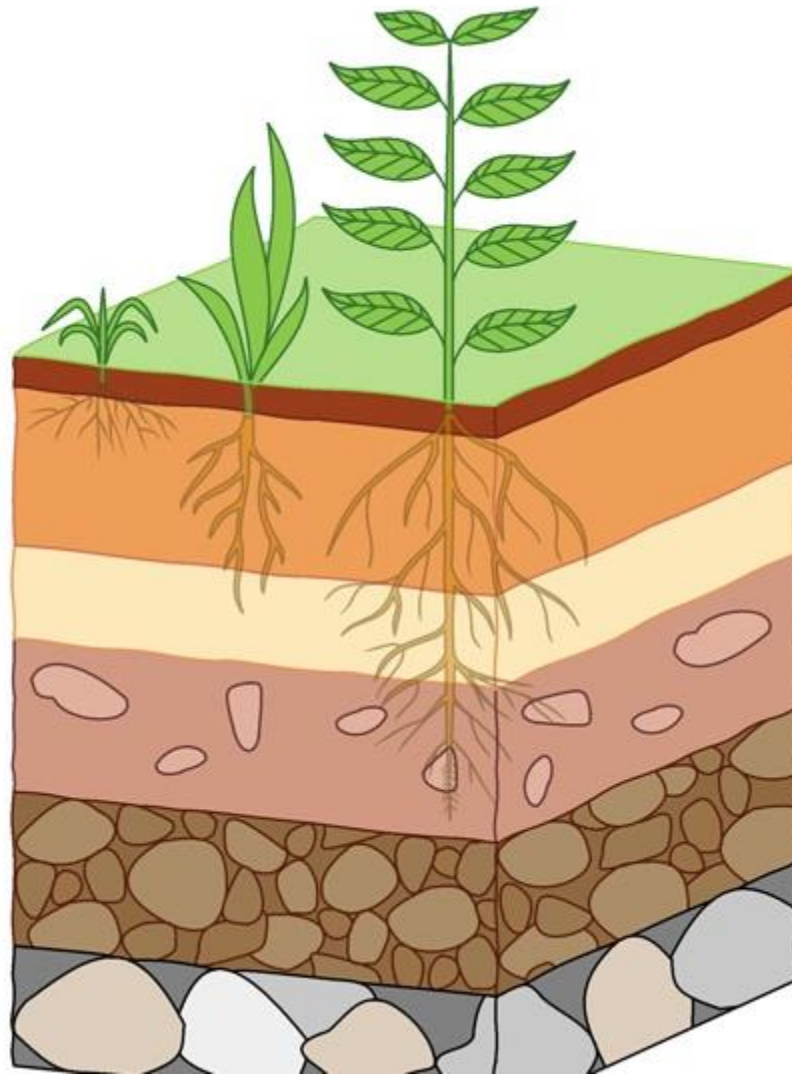
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Soil Horizon

O - Organic Layer
Humus

A - Topsoil
Minerals with humus

E - Eluviation Layer
Leached minerals & organic matter

B - Subsoil
Deposited minerals & metal salts

C - Parent Rock
Partly weathered rock

R - Bedrock
Unweathered parent rock

92. This is determined by placing an air-dry soil in a nearly saturated atmosphere at 25°C until it absorbs no more water. This tension is equal to a force of 31 atm. Water at this tension is not available to plants.

- a. field capacity water
- b. hygroscopic coefficient
- c. gravitational coefficient
- d. capillary coefficient

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93. A plant that thrives under medium conditions of moisture and salt content of the soil.

a.mesophyte

b.sporophyte

c.thermophyte

d.agrophyte

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94. A farmer has 2,400 feet of fencing and wants to fence off a rectangular field that border a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

a. 60 by 120

b. 600 by 1,200

c. 24 by 100

d. 10 by 240

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95. A triangular plot of land is to be used as a vegetable garden. The measures of the plot are $AB = 8$ meters, $BC = 12$ meters and $AC = 18$ meters. Find the measure of angle B.

a. 127.2°

b. 20.7°

c. 90°

d. 70.2°

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a. 127.2°

Apply cosine law

b. 20.7°

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96. The mass of radium-226 is given by $m(t) = be^{-kt}$. Its half-life is 1,590 years. A sample radium-226 has a mass of 1,000 mg. Find the mass after 1,000 years.

a. 650 mg

b. 100 mg

c. 165 mg

d. 200 mg

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97. The selling price P of an item is $100 - 0.02x$ pesos, where x is the number of items produced per day. If the cost C of producing and selling x items is $40x + 15,000$, pesos per day, how many items should be produced and sold every day in order to maximize the profit?

- a. 1,000
- b. 1,250
- c. 1,500
- d. 2,000

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- c. 1,500**
- d. 2,000

$$\text{Profit} = \text{Revenue} - \text{Cost}$$

$$\text{Revenue} = (100 - 0.02x)x$$

$$\text{Cost} = 40x + 15000$$

$$\text{Equation: } 100x - 0.02x^2 - 40x - 15000 = 0 \quad (\text{derive})$$

$$100 - 0.04x - 40 = 0$$

$$x = 1500$$

98. A ladder 10 feet long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate of 1 ft/sec, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 6 feet from the wall?

a. 0.75 ft/sec

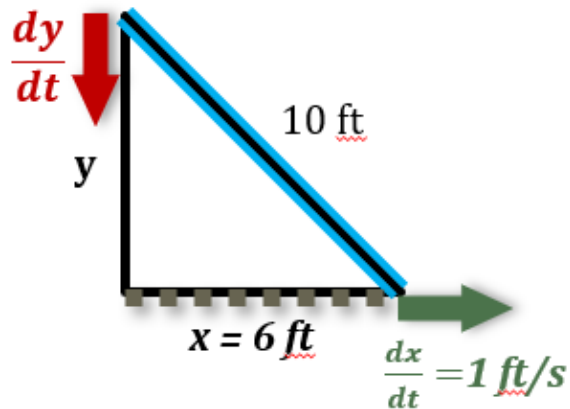
b. -0.75 ft/sec

c. 0.5 ft/sec

d. -0.5 ft/sec

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- b. -0.75 ft/sec**
- c. 0.5 ft/sec
- d. -0.5 ft/sec



By Pythagorean Theorem,

$$x^2 + y^2 = 10^2$$

Differentiating both sides wrt time,

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$x \frac{dx}{dt} + y \frac{dy}{dt} = 0$$

Getting the value of y:

$$x^2 + y^2 = 10^2$$

$$y = \sqrt{10^2 - 6^2} = 8 \text{ ft}$$

$$x \frac{dx}{dt} + y \frac{dy}{dt} = 0$$

$$6(1) + 8 \frac{dy}{dt} = 0$$

$$8 \frac{dy}{dt} = -6$$

$$\frac{dy}{dt} = -0.75 \text{ ft/s}$$

99. A water tank has the shape of an inverted circular cone with the base radius 2 meters and height 4 meters. If water is being pumped into the tank at a rate of $2 \text{ m}^3/\text{minute}$, find the rate at which the water level is rising when the water is 3 meters deep.

- a. $-0.28/\text{min}$
- b. 0.28 m/min
- c. 0.4m/min
- d. -0.4 m/min

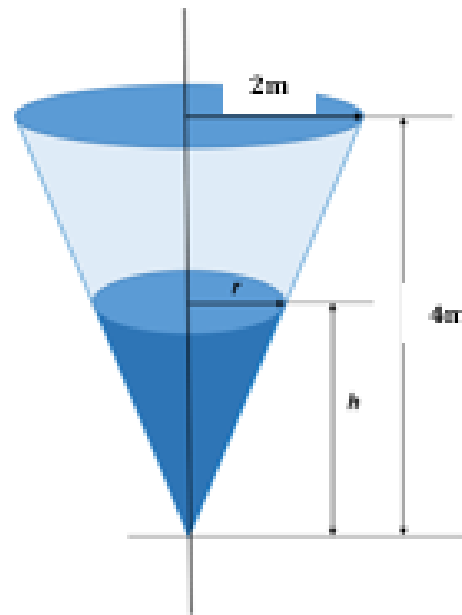
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$$\frac{r}{h} = \frac{R}{H}$$

$$\frac{r}{h} = \frac{2}{4}$$

$$r = \frac{h}{2}$$

$$V = \frac{1}{3} \pi r^2 h$$

$$V = \frac{1}{3} \pi \left(\frac{h}{2} \right)^2 h$$

$$V = \frac{1}{12} \pi h^3$$

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$$\frac{dV}{dt} = \left(\frac{1}{12} \pi \right) (3h^2) \frac{dh}{dt}$$

$$\frac{dV}{dt} = \frac{1}{4} \pi h^2 \frac{dh}{dt}$$

$$2 = \frac{1}{4} \pi (3)^2 \frac{dh}{dt}$$

$$\frac{dh}{dt} = 0.28 \text{ m/min}$$

100. A pond bottom slope of 2000:1 is equivalent to:

a. 0.05 % slope

b. 0.5 % slope

c. 1.0 % slope

d. 5 % slope

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